

REPRODUCIBLE METHODS

Data Source

The data used in this analysis were obtained from the Safe Drinking Water Information System (SDWIS) Federal Reporting Services database, maintained by the U.S. Environmental Protection Agency (EPA). The dataset includes reported drinking water violations for public water systems across the United States. For this project, violation data were collected for five states representing different EPA regions: Georgia, Texas, Ohio, Massachusetts, and Washington. Only violations associated with the Total Coliform Rule (TCR) were considered, as this rule is used as an indicator of microbial contamination and overall system integrity.

Data Collection

Data was collected directly from the SDWIS online reporting interface by generating violation reports for each of the five selected states. The report type “Violations” was selected, and filters were applied to ensure consistency across all datasets. Specifically, the following parameters were used:

- Submission Year: 2025
- Quarter: 4
- Report Type: Violations
- Water System Status: Active systems only

Additional filters were left as default was downloaded individually and exported for further processing. This step ensured that all data used in the analysis were standardized across states.

Data Cleaning and Preparation

The raw SDWIS data were cleaned and prepared using Microsoft Excel prior to analysis. Unnecessary columns were removed to streamline the dataset, and relevant variables were retained, including:

- State
- Public Water System (PWS) ID
- Population served
- Primary water source (groundwater or surface water)
- Violation type
- Compliance period dates

Date fields were reformatted to ensure consistency, and any incomplete or duplicate records were reviewed and removed as needed. After cleaning, the finalized dataset was exported as a comma-separated values (CSV) files for compatibility with Python-based analysis tools.

Data Processing

The cleaned CSV dataset was imported into Python for analysis using the pandas and numpy libraries. Pandas was used for data manipulation, including grouping, filtering, and aggregation, while numpy supported numerical operations. Custom functions were defined to automate repetitive tasks such as grouping violations by category and calculating summary statistics. This ensured consistency and reproducibility across all analyses.

Data Analysis

Two primary analyses were conducted to address the client's research questions.

First, TCR violations were grouped by state and year to evaluate temporal trends and identify differences in violation frequency across the five states. Aggregated counts of violations were calculated for each state to allow for direct comparison.

Second, violations were analyzed based on system size to determine whether smaller water systems experience more TCR violations than larger systems. System size categories were defined using population served. Systems were grouped into size categories, and total violations within each category were calculated and compared.

These analyses were designed to identify both spatial (state-level) and structural (system size) in TCR violations.

Data Visualization

Results from the analysis were visualized using Python plotting libraries. Summary tables and graphical outputs were created to clearly communicate trends and comparisons. At minimum, two types of visualizations were generated:

- A time-based plot showing TCR violations by state
- A comparative plot showing violations by system size category

These visualizations were used to support interpretation and highlight key patterns in the data.

Interpretation and Supporting Research

Following the quantitative analysis, results were interpreted in the context of existing literature and regulatory guidance. External sources, including EPA documentation on the Revised Total Coliform Rule (RTCR) and peer-reviewed studies on drinking water compliance, were reviewed to explain observed differences. Factors such as monitoring frequency, infrastructure condition, regulatory variation between states, and the operational challenges of small or seasonal systems were considered when interpreting results.

Reproducibility

All steps in this analysis, from data collection through visualization, were documented to ensure reproducibility. The workflow includes:

1. Accessing SDWIS and applying specified filters
2. Exporting violation data for each state
3. Cleaning and formatting data in Excel
4. Importing the cleaned dataset into Python
5. Performing analysis using pandas and numpy
6. Generating tables and visualizations